GEOMETRIC MEASURES FOR RADIAL MINKOWSKI HOMOMORPHISMS AND BRUNN-MINKOWSKI-TYPE INEQUALITIES

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Abstract. For a radial Minkowski homomorphism Φ, we establish an Aleksandrov-type variational formula for $V(\Phi K)$ and introduce the associated geometric measures. We also obtain existence and uniqueness results for the corresponding Minkowski problem. Finally, we derive a new Brunn-Minkowski-type inequality.

1. Introduction

A central goal in convex geometry is to identify geometric invariants and measures that can be used to determine the shape of convex bodies. Using geometric invariants such as volume, surface area, or more generally quermassintegrals and dual quermassintegrals, special convex bodies can be characterized as extremals of isoperimetric inequalities. On the other hand, to reconstruct a convex body of arbitrary shape, different notions of curvatures, or, in the absence of smoothness, geometric measures, are necessary. At the turn of the last century, Minkowski [43] used *surface area measure* to reconstruct polytopes. Aleksandrov [1] revealed that the surface area measure appearing in the Minkowski problem can be obtained by differentiating the volumes of a properly parametrized family of convex bodies: if K is a convex body and $f \in C(S^{n-1})$, then

$$\frac{d}{dt}\bigg|_{t=0} V(K_t) = \int_{S^{n-1}} f(v) dS(K, v), \tag{1.1}$$

where $S(K, \cdot)$ is the surface area measure of K, $K_t = [h_K + tf]$ is the *Wulff shape* generated by $h_K + tf$, and h_K is the support function of K. Here and throughout, classical notions not explained explicitly can be found in Section 2.

In fact, assuming smoothness[1], the Minkowski problem is known as the problem of prescribing Gauss curvature. Owing to Minkowski, Aleksandrov, Fenchel, and Jessen, it is now well known that the surface area measure (or Gauss curvature) can be used to reconstruct and *uniquely identify* the shape of *every* convex body. Among the many influential works in the last century are those of Cheng-Yau [13] and Pogorelov [44]. Towards the end of the last century, Caffarelli [6–8] showed that one can infer the regularity of a convex body by simply studying the regularity of its surface area measure. Over the past two decades, inspired by isoperimetric inequalities involving many other invariants, other important geometric measures and Minkowski problems have been intensively studied. See the recent survey by Huang-Yang-Zhang [28]. Each of these problems is associated with a distinct, possibly degenerate Monge-Ampère-type equation on the unit sphere.

2020 *Mathematics Subject Classification.* 52A40, 52A20, 35J96.

Key words and phrases. Aleksandrov variation formula, radial Minkowski homomorphism, Minkowski problem, Monge-Ampère equations, Brunn-Minkowski inequality.

[1] which is not a requirement in the Minkowski problem

Let $\mathcal{K}_o^n$ be the set of convex bodies containing the origin in their interiors and $\mathcal{S}_o^n$ be the set of star bodies equipped with the usual topology. Let $\Phi : \mathcal{A} \to \mathcal{G}$ be a map where $\mathcal{A}$ and $\mathcal{G}$ are either $\mathcal{K}_o^n$ or $\mathcal{S}_o^n$. It is natural to ask whether information obtained from ΦQ can be used to extract data on Q. To give an example, the celebrated Busemann-Petty problem asks whether $IK \subset IQ$ for a pair of origin-symmetric convex bodies K and Q necessarily implies $V(K) \leq V(Q)$. Here I is the *intersection body* operator. Among other classical examples of Φ, one may count the projection body operator, the centroid body operator, the intersection body operator, the polar body operator, and the difference body operator. They are often counterparts of well-known transformations—such as spherical cosine or Radon transforms—in analysis.

The volume $V(\Phi K)$ for different Φ provides a rich library of geometric invariants. Many classical isoperimetric inequalities and their reverses are inequalities involving $V(\Phi K)$ for different Φ's. The Busemann-Petty inequality involving the centroid body operator and its L_p dual by Lutwak-Zhang [41], the Petty projection inequality involving the polar projection body operator and its reverse, the Zhang projection inequality [55], Petty's conjecture involving the projection body operator, the Busemann intersection inequality involving the intersection body operator, and the Rogers-Shephard inequality involving the difference body operator are just a few such examples.

The current work seeks answers to the following three basic questions.

Q1. Does $V(\Phi K)$ satisfy a Brunn-Minkowski type inequality?

Q2. Let K_t be a perturbation of K. How does $V(\Phi K_t)$ change?

The answer to Q2 is known as an Aleksandrov-type variational formula. See (1.1). This often takes the following form:

$$\frac{d}{dt}\bigg|_{t=0} V(\Phi K_t) = \int_{S^{n-1}} f(v) dS_\Phi(K, v),$$

for some geometric measure $S_\Phi(K, \cdot)$ that depends on both Φ and K. Thus, a positive answer to Q2 naturally leads to the following question.

Q3. Given a Borel measure ν on the unit sphere, can one find $K \in \mathcal{K}_o^n$ such that $S_\Phi(K, \cdot) = \nu$? Is the solution unique?

Q3 is known as a Minkowski-type problem. See Section 4 for more detail. Compare it to the classical Minkowski problem of surface area measures.

Note that if Φ is the trivial identity map, then Q1–Q3 are simply the classical Brunn-Minkowski inequality, the Aleksandrov variational formula, and the classical Minkowski problem, respectively.

Naturally, a one-size-fits-all answer to these questions is unlikely to exist. They have been studied in various special cases. When Φ is a Blaschke Minkowski homomorphism or its polar, Q1 was answered completely in Schuster [49]; when Φ is a radial Blaschke-Minkowski homomorphism, *a variant* of Q1—a *dual* Brunn-Minkowski-type inequality—was shown by Schuster [49][2]; when Φ is the polar projection body operator or the radial mean body operator, Q2 and Q3 were partially answered in Xi-Zhao [53]; when Φ is the intersection body operator and its L_p variant, Q2 and Q3 were treated in Cai-Leng-Wu-Xi [9] and most recently in Lin-Wu [33].

[2]Note that while the addition on the left side of Theorem 1.5_{d} is the Minkowski sum, it is a typo. It should have been the dual Minkowski sum. See Theorem 7.6, which relies on first using (7.3) followed by (3.11). Inequality (3.11) demands that the sum is the dual Minkowski sum.

Note that while the map Φ might be defined on $\mathcal{S}_o^n$, for the purpose of Q2 and Q3, it is often necessary to restrict to $\mathcal{K}_o^n$, as answers to them often involve the Gauss map and thus require convexity.

In the current work, we answer Q1–Q3 for a family of Φ, known as *radial Minkowski homomorphisms.*

For a star body K, we denote its *radial function* by ρ_K. See (2.1) for its precise definition. If L is another star body and $p \neq 0$, we use $K \widetilde{+}_p L$ to denote the dual L_p Minkowski sum of K and L. See (2.5).

Definition 1.1. *Let $r > 0$ and $q \neq 0$. A map $\Phi : \mathcal{S}_o^n \to \mathcal{S}_o^n$ is an (r, q) radial Minkowski homomorphism if*

(1) Φ is continuous;

(2) Φ intertwines $\mathrm{SO}(n)$: $\Phi(\vartheta K) = \vartheta \Phi(K)$ for all $\vartheta \in \mathrm{SO}(n)$ and $K \in \mathcal{S}_o^n$;

(3) for every $K, L \in \mathcal{S}_o^n$, we have

$$\Phi(K \widetilde{+}_r L) = \Phi(K) \widetilde{+}_q \Phi(L);$$

(4) As $c \to 0^+$, we have

$$\rho_{\Phi(cB)} \to \begin{cases} 0, & \text{if } q > 0, \\ +\infty, & \text{if } q < 0, \end{cases}$$

uniformly on S^{n-1}. Here, B is the centered unit ball in $\mathbb{R}^n$.

In particular, an $(n-1, 1)$ radial Minkowski homomorphism is also known as a *radial Blaschke-Minkowski homomorphism.* Radial Blaschke-Minkowski homomorphisms, together with their associated isoperimetric inequalities and mixed volumes were studied extensively in Schuster [49].

Classical examples of radial Minkowski homomorphisms include the intersection body operator introduced by Lutwak [36], its L_p and asymmetric variants (see, for example, Haberl [23], and Haberl-Ludwig [24]), and the L_p polar centroid body operator (see, for example, Gardner-Giannopoulos [19], Lutwak-Zhang [41] and Yaskin-Yaskina [54]). It is well-known that radial Minkowski homomorphisms are intimately connected to spherical convolutions, and there is, in fact, an abundant family of such maps beyond the classical operators mentioned above. See Theorem 2.3 and Remark 2.4.

We remark that the assumption Φ is continuous is essential in answering Q2 since otherwise $V(\Phi K_t)$ might be discontinuous, let alone differentiable.

The following theorem provides a positive answer to Q2.

Theorem 1.2. *Let $r > 0, q \neq 0$, $p \in \mathbb{R}$, $\Omega \subset S^{n-1}$ be a closed set not contained in any closed hemisphere. Suppose $h_0, f : \Omega \to (0, \infty)$ are continuous functions. For sufficiently small $|t|$, define $h_t : \Omega \to (0, \infty)$ by*

$$\begin{aligned} h_t^p &= h_0^p + tf + o(t, \cdot), && \text{if } p \neq 0, \\ \log h_t &= \log h_0 + tf + o(t, \cdot), && \text{if } p = 0, \end{aligned}$$

where $o(t, \cdot)$ is a continuous function on Ω and $o(t, \cdot)/t$, as $t \to 0$, converges uniformly to 0 on Ω. Write $K_t = [h_t]$. If $\Phi : \mathcal{S}_o^n \to \mathcal{S}_o^n$ is an (r, q) radial Minkowski homomorphism, then

$$\left.\frac{d}{dt}\right|_{t=0} V(\Phi K_t) = \begin{cases} \frac{r}{pq} \int_\Omega f(v) dS_{\Phi,p}(K, v), & \text{if } p \neq 0, \\ \frac{r}{q} \int_\Omega f(v) dS_{\Phi,0}(K, v), & \text{if } p = 0. \end{cases}$$

Here, the measure $S_{\Phi,p}(K,\cdot)$ is called the L_p Φ-*surface area measure* of $K\in\mathcal{K}_o^n$: it is a finite Borel measure on S^{n-1} given by

$$S_{\Phi,p}(K,\omega)=\int_{\boldsymbol{\alpha}_K^*(\omega)}(\rho_K(u)u\cdot\alpha_K(u))^{-p}\rho^q_{\Phi\mathcal{T}\Phi K}(u)\rho_K^r(u)du,$$

where $\omega\subset S^{n-1}$ is a Borel subset, $\boldsymbol{\alpha}_K^*$ is the inverse radial Gauss image, and $\mathcal{T}=\mathcal{T}_{r,q}:\mathcal{S}_o^n\to\mathcal{S}_o^n$ is the re-normalization map given by

$$\rho_{\mathcal{T}_{r,q}K}(u)=\rho_K^{\frac{n-q}{r}}(u),\ \text{for each } u\in S^{n-1}.$$

The family of L_p Φ-surface area measures is rich and includes many classical and well-studied examples.

- In the extremely special case, when Φ is the identity map, $S_{\Phi,p}(K,\cdot)$ recovers the classical L_p surface area measure introduced in the landmark work [37,38] by Lutwak.
- For a *fixed* $Q\in\mathcal{S}_o^n$ and $s\neq 0$, consider the renormalization map:

$$\rho_{\Phi K}(u)=\rho_K^{\frac{s}{n}}(u)\rho_Q^{\frac{n-s}{n}}(u),\ \text{for each } u\in S^{n-1},\tag{1.2}$$

  the volume $V(\Phi K)$ recovers the dual mixed volume of K and Q. Consequently, as special cases, $S_{\Phi,p}(K,\cdot)$ recovers
  - the dual curvature measures introduced in the pioneering work [26] by Huang-Lutwak-Yang-Zhang;
  - the L_p dual curvature measures introduced in Lutwak-Yang-Zhang [40].
- When Φ is the intersection body operator, or, more generally its L_p extension, defined as

$$\rho^p_{\Phi K}(u)=\frac{1}{\Gamma(1-p)}\int_K|x\cdot u|^{-p}dx \text{ for each } u\in S^{n-1},$$

  where Γ is the Gamma function and $0\neq p<1$, the family of $S_{\Phi,p}(K,\omega)$ includes the recently introduced $(n-1)$-th affine dual curvature measure by Cai-Leng-Wu-Xi [9] and its p-version in Lin-Wu [33].

We remark once again that since there are infinitely many more radial Minkowski homomorphisms, the family of L_p Φ-surface area measures encompasses much more than the existing ones mentioned above.

In the past few decades, the study of these geometric measures has occupied much of the research in convex geometric analysis and fully nonlinear elliptic PDEs (particularly those of Monge-Ampère type).

For example, the L_p surface area measure introduced by Lutwak [37,38] is a part of the fruitful L_p Brunn-Minkowski theory. In particular, the theory becomes *significantly* harder when $p<1$. Isoperimetric inequalities and Minkowski problems in neither case have been fully addressed. The singular $p=0$ is particularly important. The conjectured inequality, in this case, is known as the *log Brunn-Minkowski conjecture* (see Böröczky-Lutwak-Yang-Zhang [4]). It is arguably one of the most crucial conjectures in convex geometric analysis in the past decade. The log Brunn-Minkowski conjecture has been verified in dimension two and in various special classes of convex bodies. See, for example, Chen-Huang-Li-Liu [11], Colesanti-Livshyts-Marsiglietti [15], Kolesnikov-Livshyts [30], Kolesnikov-Milman [29], Milman [42], Putterman [45], Saroglou [47], and Xi-Leng [51]. If proven correct, it is much stronger than the classical Brunn-Minkowski inequality. The corresponding Minkowski problem is the *log Minkowski problem* for the *cone volume measure*, which is not fully resolved either. Partial solutions in the $p \leq 0$ case can be found in Bianchi-Böröczky-Colesanti-Yang [2], Böröczky-Lutwak-Yang-Zhang [5], Chou-Wang [14], Guang-Li-Wang [22], Zhu [57, 58] among many other works. More recently, these measures and their corresponding Minkowski problems have been extended to $\mathbb{R}^n$ with non-uniform background measures. See, for example, Livshyts [35], Fradelizi-Langharst-Madiman-Zvavitch [17], Huang-Xi-Zhao [27], and Kryvonos-Langharst [31].

As another example, the (L_p) dual curvature measures introduced in Huang-Lutwak-Yang-Zhang [26] and subsequently [40] are dual to Federer's curvature measures. They are the long-sought-for counterparts in the dual Brunn-Minkowski theory introduced by Lutwak in the 1970s. The Minkowski problem for dual curvature measures, now known as the *dual Minkowski problem*, has been central in convex geometry and related fully nonlinear elliptic PDEs for the last couple of years and has already led to a number of papers in a short period. See, for example, Böröczky-Henk-Pollehn [3], Chen-Huang-Zhao [10], Chen-Li [12], Gardner-Hug-Weil-Xing-Ye [20], Henk-Pollehn [25], Li-Sheng-Wang [32], Liu-Lu [34], Zhao [56].

Note that all of those previously individually discovered and treated geometric measures are, as a matter of fact, special cases of Φ-surface area measures. It is surprising that Q3 can be treated in a uniform way for all of them. For easier future reference, we will refer to Q3 for $S_{\Phi,p}$ as the L_p *Φ-Minkowski problem*.

It is simple to see from Theorem 2.3 that if Φ is an (r, q) radial Minkowski homomorphism, then Φ is monotone; that is, if $K, L \in \mathcal{S}_o^n$ are such that $K \subset L$, then $\Phi K \subset \Phi L$ when $q > 0$ and $\Phi K \supset \Phi L$ when $q < 0$. We say that Φ is *strictly monotone* if equality holds in the above inclusion only when $K = L$.

Theorem 1.3. *Let $r, p > 0$, $q \neq 0$ with $nr \neq pq$, $\Phi : \mathcal{S}_o^n \to \mathcal{S}_o^n$ be an (r, q) radial Minkowski homomorphism, and ν be a nonzero, finite, even Borel measure on S^{n-1}. There exists $K_0 \in \mathcal{K}_e^n$ such that $S_{\Phi,p}(K_0, \cdot) = \nu$ if and only if ν is not concentrated in any great subsphere.*

If, in addition, $r < p$, Φ is strictly monotone, and either

- *$q < 0$, or,*
- *$q > 0$ and $nr < pq$,*

the solution K_0 is unique.

Theorem 1.3 follows from Theorems 4.5 and 4.7.

It is worth pointing out that when $r = n - q$, the L_p Φ-surface area measure is related in the following way to the L_p dual curvature measure $\widetilde{C}_{p,q}(K, Q)$ introduced by Lutwak-Yang-Zhang [40]:

$$S_{\Phi,p}(K, \cdot) = n\widetilde{C}_{p,n-q}(K, \Phi\mathcal{T}\Phi K, \cdot).$$

This realization could lead to the **misunderstanding** that the L_p Φ-Minkowski problem

$$\nu = S_{\Phi,p}(K, \cdot) = n\widetilde{C}_{p,n-q}(K, \Phi\mathcal{T}\Phi K, \cdot), \tag{1.3}$$

is similar in nature to the L_p dual Minkowski problem

$$\nu = \widetilde{C}_{p,n-q}(K, Q, \cdot). \tag{1.4}$$

To see that this is indeed *not* true, note that the bodies K and Q in (1.4) are *independent* of each other while the measure equation (1.3) demands that $Q = \Phi\mathcal{T}\Phi K$ (in other words, more restrictive).

Another way to see the *drastically different* natures of (1.3) and (1.4) is to look at the underlying Monge-Ampère equations.

With sufficient regularity, (1.3) reduces to the following Monge-Ampère equation on the unit sphere:

$$h_K^{1-p}(v)\rho^q_{\Phi\mathcal{T}\Phi K}(\alpha^*_K(v))|\nabla h_K(v)+h_K(v)v|^{r-n}\det(\nabla^2 h_K(v)+h_K(v)I)=f(v). \tag{1.5}$$

See (2.2) and the paragraph following it for the precise definition of α^*_K. Here, ∇ and ∇^2 are the gradient and Hessian on S^{n-1}, respectively. The PDE (1.5) reveals that unlike (1.4), the measure equation (1.3) is *non-local*; *i.e.,* knowing the shape of ∂K in a small neighborhood is not sufficient to determine whether h_K satisfies (1.5) in a small neighborhood. The problem lies in the *global* term $\rho_{\Phi\mathcal{T}\Phi K}$ appearing in the equation. This distinct feature makes the PDE (1.5) quite challenging[3]. This *non-local* feature first appeared in the chord Minkowski problem studied in the groundbreaking work [39] by Lutwak-Xi-Yang-Zhang.

The uniqueness part in Theorem 1.3 is established via a positive answer to Q1.

Theorem 1.4. *Let $0<r<p$, $q\neq 0$, and $\Phi:\mathcal{S}^n_o\to\mathcal{S}^n_o$ be an (r,q) radial Minkowski homomorphism. Assume $q<0$, or else $q>0$ and $nr\le pq$. Then for every $K,L\in\mathcal{K}^n_o$ and $\lambda\in(0,1)$, we have*

$$V(\Phi((1-\lambda)\cdot K+_p\lambda\cdot L))^{\frac{pq}{rn}}\ge(1-\lambda)V(\Phi K)^{\frac{pq}{rn}}+\lambda V(\Phi L)^{\frac{pq}{rn}}. \tag{1.6}$$

Moreover, if Φ is strictly monotone, then equality holds in (1.6) *if and only if K and L are dilates of each other.*

Open problem. Does (1.6) or (1.6) with inequality reversed hold for other p,q,r?

Note that when Φ is the identity map, (1.6) becomes the well-known Brunn-Minkowski inequality and its L_p version. See the beautiful survey [18] by Gardner. When Φ is the renormalization map given in (1.2) with $K=B$ and K,L are origin-symmetric, (1.6) is a conjecture of Lutwak's regarding s-th dual quermassintegrals and has very recently been established in Sadovsky-Zhang [46] for $s\in(0,n]$. For $s\le 1$, this was established in Xi-Zhang [52] together with a characterization of the equality condition.

2. Preliminaries

This section contains an overview of the notions and results needed in the current work.

2.1. **Basics of convex bodies.**

We collect here some basic facts about convex bodies. The comprehensive book [48] by Schneider is a good complement to what is discussed here.

Throughout this work, we work in Euclidean space $\mathbb{R}^n$ with $n\ge 2$. The standard inner product and norm are denoted by $x\cdot y$ and $|x|$ for $x,y\in\mathbb{R}^n$, respectively. We write B for the centered unit ball and S^{n-1} for its boundary, the unit sphere. The volume of the unit ball B is denoted by ω_n. The space $C(S^{n-1})$ of all continuous functions on S^{n-1} is equipped with the standard supremum norm; *i.e.*, $\|f\|_\infty=\max\{|f(u)|:u\in S^{n-1}\}$. We will use the superscript $+$ for positive functions and the subscript e for even functions; that is, for example, the subspace $C^+_e(S^{n-1})\subset C(S^{n-1})$ contains all positive even continuous functions on S^{n-1}. For each $u\in S^{n-1}$, we will write $u^\perp$ for the $(n-1)$-dimensional subspace orthogonal to u.

A set $K\subset\mathbb{R}^n$ is called a *convex body* if it is a compact convex set with nonempty interior. In this work, we focus on those that contain the origin as an interior point; this collection is

[3]and provides a clear indication that the L_p Φ-Minkowski problem is vastly different from the L_p dual Minkowski problem in [40].

denoted by $\mathcal{K}_o^n$. The subcollection of $\mathcal{K}_o^n$ consisting of origin-symmetric bodies will be denoted by $\mathcal{K}_e^n$. For a compact convex set $K \subset \mathbb{R}^n$, its support function $h_K : S^{n-1} \to \mathbb{R}$ is given by

$$h_K(v) = \max_{x \in K} x \cdot v, \text{ for each } v \in S^{n-1}.$$

A compact set $Q \subset \mathbb{R}^n$ is a *star-shaped* set with respect to the origin if $o \in Q$ and the intersection of any ray emanating from o with Q is always a line segment. Its *radial function* $\rho_Q : S^{n-1} \to \mathbb{R}$ is defined by

$$\rho_Q(u) = \max\{t \geq 0 : tu \in Q\}. \tag{2.1}$$

If ρ_Q is a positive continuous function, we say Q is a *star body*. The collection of all star bodies in $\mathbb{R}^n$ is denoted by $\mathcal{S}_o^n$. It is simple to see that $\mathcal{K}_o^n \subset \mathcal{S}_o^n$.

For simplicity in various settings, we often extend h_K to $\mathbb{R}^n$ as a homogeneous function of degree 1 and ρ_K to $\mathbb{R}^n \setminus \{o\}$ as a homogeneous function of degree -1.

Let K_1, K_2 be two compact convex subsets of $\mathbb{R}^n$. Their *Hausdorff distance* is given by

$$d_H(K_1, K_2) = \|h_{K_1} - h_{K_2}\|_\infty.$$

The set $\mathcal{K}_o^n$ is equipped with the topology induced by this metric. Similarly, let Q_1, Q_2 be two star-shaped sets whose radial functions are continuous. We define the *radial distance* by

$$\widetilde{d_H}(Q_1, Q_2) = \|\rho_{Q_1} - \rho_{Q_2}\|_\infty.$$

The set $\mathcal{S}_o^n$ is equipped with the topology induced by this metric. It is simple to see that if $K_i, K \in \mathcal{K}_o^n$, then $K_i \to K$ in Hausdorff metric if and only if $K_i \to K$ in radial metric.

Let $\Omega \subset S^{n-1}$ be a closed set not contained in any closed hemisphere. For each $f \in C(\Omega)$, define $[f, \Omega]$ by

$$[f, \Omega] = \bigcap_{v \in \Omega} \{x \in \mathbb{R}^n : v \cdot x \leq f(v)\}.$$

When $[f, \Omega]$ is nonempty, it is called the *Wulff shape* or the *Aleksandrov body* generated by f. It is simple to see

$$h_{[f,\Omega]} \leq f \text{ on } \Omega,$$

and if $K \in \mathcal{K}^n$, then

$$[h_K, S^{n-1}] = K.$$

When the context is clear, we will simply write $[f]$ for $[f, \Omega]$.

For each $K \in \mathcal{K}_o^n$ and $\eta \subset \partial K$, the *Gauss image* $\boldsymbol{\nu}_K(\eta)$ is given by

$$\boldsymbol{\nu}_K(\eta) = \{v \in S^{n-1} : v \cdot z = h_K(v) \text{ for some } z \in \eta\}.$$

Its radial normalized version, the *radial Gauss image* $\boldsymbol{\alpha}_K(\sigma)$, for each $\sigma \subset S^{n-1}$, is given by

$$\boldsymbol{\alpha}_K(\sigma) = \boldsymbol{\nu}_K(\{\rho_K(u)u : u \in \sigma\}).$$

At each boundary point $z \in \partial K$, a unit vector $v \in S^{n-1}$ is called an *outer unit normal* of K if $v \in \boldsymbol{\nu}_K(z)$. By convexity, for $\mathcal{H}^{n-1}$-almost every $z \in \partial K$, the set $\boldsymbol{\nu}_K(\{z\})$ is a singleton, and in this case, we will denote the unique outer unit normal at z by $\nu_K(z)$. (Such a property cannot be expected for bodies in $\mathcal{S}_o^n$.) We will similarly write $\alpha_K(u)$ when $\boldsymbol{\alpha}_K(\{u\})$ is a singleton. Note that this holds for $\mathcal{H}^{n-1}$-almost every $u \in S^{n-1}$.

The *inverse Gauss image*, $\boldsymbol{\nu}_K^*(\omega)$, for each Borel set $\omega \subset S^{n-1}$, is given by

$$\boldsymbol{\nu}_K^*(\omega) = \{z \in \partial K : v \cdot z = h_K(v) \text{ for some } v \in \omega\}.$$

Its radial normalized version, the *inverse radial Gauss image*, $\boldsymbol{\alpha}_K^*(\omega)$, is given by

$$\boldsymbol{\alpha}_K^*(\omega) = \{u \in S^{n-1} : \rho_K(u)u \in \boldsymbol{\nu}_K^*(\omega)\}. \tag{2.2}$$

When $\boldsymbol{\nu}_K^*(\{v\})$ or $\boldsymbol{\alpha}_K^*(\{v\})$ is a singleton for some $v \in S^{n-1}$, we will write $\nu_K^*(v)$ or $\alpha_K^*(v)$, respectively. Note that this holds for $\mathcal{H}^{n-1}$-almost every $v \in S^{n-1}$.

The *surface area measure* of K appearing in (1.1), perhaps the most studied geometric measure in convex geometric analysis, is given by

$$S(K,\omega) = \mathcal{H}^{n-1}(\boldsymbol{\nu}_K^*(\omega)),$$

for every Borel set $\omega \subset S^{n-1}$, where $\mathcal{H}^{n-1}$ is the $(n-1)$-dimensional Hausdorff measure.

Let $0 \neq p \in \mathbb{R}$ and $K, L \in \mathcal{K}_o^n$. The L_p *Minkowski sum*, $K +_p L$, of K and L is given by

$$K +_p L = \left[(h_K^p + h_L^p)^{\frac{1}{p}}\right]. \tag{2.3}$$

Note that when $p \geq 1$, since $(h_K^p + h_L^p)^{\frac{1}{p}}$ is convex, (2.3) yields

$$h_{K+_pL} = (h_K^p + h_L^p)^{\frac{1}{p}}.$$

However, this is not true for $p < 1$ in general. When $p = 1$, this recovers the usual Minkowski sum. For $\lambda \geq 0$, the L_p *scalar multiplication* is given by

$$\lambda \cdot_p K = \lambda^{\frac{1}{p}} K. \tag{2.4}$$

We will often suppress the subscript p when the context is clear.

Combining (2.3) and (2.4), we obtain

$$(1-\lambda) \cdot K +_p \lambda \cdot L = \left[((1-\lambda)h_K^p + \lambda h_L^p)^{\frac{1}{p}}\right],$$

for every $\lambda \in [0,1]$. This formulation allows us to extend (2.3):

$$(1-\lambda) \cdot K +_0 \lambda \cdot L = \left[h_K^{1-\lambda} h_L^{\lambda}\right].$$

The L_p Minkowski combination when $p > 1$ was initially introduced by Firey and a systematic investigation was started by Lutwak [37] in the 1990s, which marks the beginning of the now fruitful L_p Brunn-Minkowski theory.

Similarly, for $0 \neq p \in \mathbb{R}$ and $K, L \in \mathcal{S}_o^n$, the *dual* L_p *Minkowski sum*, $K \widetilde{+}_p L$, is the star body whose radial function is given by

$$\rho_{K\widetilde{+}_pL} = (\rho_K^p + \rho_L^p)^{\frac{1}{p}}. \tag{2.5}$$

In particular, for $\lambda \in [0,1]$, we have

$$\rho_{(1-\lambda)\cdot K\widetilde{+}_p\lambda\cdot L} = ((1-\lambda)\rho_K^p + \lambda\rho_L^p)^{\frac{1}{p}}.$$

The L_p linear combination and its dual satisfy the following relation, see [52, Lemma 3.1]: for all $K, L \in \mathcal{K}_o^n$ and $p \in \mathbb{R}, \lambda \in (0,1)$,

$$(1-\lambda) \cdot K +_p \lambda \cdot L \supset (1-\lambda) \cdot K \widetilde{+}_p \lambda \cdot L, \text{ for each } \lambda \in (0,1), \tag{2.6}$$

with equality if and only if K and L are dilates.

2.2. **Spherical convolution.** In this subsection, we discuss basic facts concerning convolutions on the sphere. For more detailed discussion of the results referenced here, Grinberg-Zhang [21, Sections 2 and 3] and Schuster [50, Section 2] are recommended.

Throughout this work, fix a point $\widehat{e} \in S^{n-1}$. Let $\mathrm{SO}(n)$ be the group of rotations on $\mathbb{R}^n$ equipped with the uniform topology, and let $\mathrm{SO}(n-1) \subset \mathrm{SO}(n)$ be the subgroup consisting of rotations that leave $\widehat{e}$ unchanged.

For every finite Borel measure μ on $\mathrm{SO}(n)$ and a continuous function $f : \mathrm{SO}(n) \to \mathbb{R}$, the *convolution* $f * \mu : \mathrm{SO}(n) \to \mathbb{R}$ is a continuous function given by

$$f * \mu(\vartheta) = \int_{\mathrm{SO}(n)} f(\vartheta\eta^{-1}) d\mu(\eta), \text{ for every } \vartheta \in \mathrm{SO}(n). \tag{2.7}$$

The unit sphere S^{n-1} and the quotient group $\mathrm{SO}(n)/\mathrm{SO}(n-1)$ can be naturally identified via

$$u = \vartheta\widehat{e} \mapsto \vartheta\,\mathrm{SO}(n-1).$$

This makes it possible to identify each continuous function (measure, resp.) on S^{n-1} with a right $\mathrm{SO}(n-1)$-invariant continuous function (measure, resp.). Thus, it makes sense to define $f * \mu \in C(S^{n-1})$ for $f \in C(S^{n-1})$ and each finite Borel measure μ on S^{n-1}. See, for example, Schuster [50, Page 5572] for a precise formula. The formula becomes much simpler if the measure μ possesses additional invariance properties. We say a finite Borel measure μ on S^{n-1} is *zonal* if μ is $\mathrm{SO}(n-1)$-invariant; that is, for each Borel set $\omega \subset S^{n-1}$ and $\vartheta \in \mathrm{SO}(n-1)$, we have

$$\mu(\omega) = \mu(\vartheta\omega).$$

For notational simplicity, we denote by $\mathcal{M}(S^{n-1}, \widehat{e})$ the set of all nonnegative, finite, zonal, Borel measures on S^{n-1}. In particular, if $f \in C(S^{n-1})$ and $\mu \in \mathcal{M}(S^{n-1}, \widehat{e})$, with the identifications discussed above, (2.7) becomes, see Schuster [50, (2.9)],

$$f * \mu(u) = \int_{S^{n-1}} f(\vartheta_u v) d\mu(v), \text{ for every } u \in S^{n-1}. \tag{2.8}$$

Here and throughout this work, we will use ϑ_u to denote a rotation such that $\vartheta_u\widehat{e} = u$. It is important to note that ϑ_u satisfying this condition is not unique. However, thanks to the fact that μ is zonal, the specific choice of ϑ_u does not affect the integral, and thus the above integration is well defined.

We will make essential use of the following formula: if $\mu \in \mathcal{M}(S^{n-1}, \widehat{e})$ and $f, g \in C(S^{n-1})$, then

$$\int_{S^{n-1}} g * \mu(u) f(u) du = \int_{S^{n-1}} g(u) f * \mu(u) du. \tag{2.9}$$

This is a special case of Schuster [50, Lemma 2.2].

Let $\mu_i, \mu_0 \in \mathcal{M}(S^{n-1}, \widehat{e})$ and $f_0 \in C(S^{n-1})$. Recall that if μ_i converges to μ_0 weakly, then

$$f * \mu_i \to f * \mu_0 \text{ uniformly on } S^{n-1}. \tag{2.10}$$

See, for example, Grinberg-Zhang [21]. We will make use of the following trivial consequence.

Lemma 2.1. *Let $f_i \in C(S^{n-1})$ be a sequence of continuous functions on S^{n-1} and $\mu_i \in \mathcal{M}(S^{n-1}, \widehat{e})$ be a sequence of measures. If $f_i \to f_0$ uniformly and $\mu_i \to \mu_0$ weakly, then*

$$f_i * \mu_i \to f_0 * \mu_0, \qquad \text{uniformly on } S^{n-1}.$$

Proof. By definition, for every $u \in S^{n-1}$, we have

$$\begin{aligned} |f_i * \mu_i(u) - f_0 * \mu_i(u)| &= \left| \int_{S^{n-1}} f_i(\vartheta_u v) d\mu_i(v) - \int_{S^{n-1}} f_0(\vartheta_u v) d\mu_i(v) \right| \\ &\leq \|f_i - f_0\|_\infty |\mu_i|. \end{aligned}$$

When combined with the triangle inequality, we have

$$\begin{aligned} |f_i * \mu_i(u) - f_0 * \mu_0(u)| &\leq |f_i * \mu_i(u) - f_0 * \mu_i(u)| + |f_0 * \mu_i(u) - f_0 * \mu_0(u)| \\ &\leq \|f_i - f_0\|_\infty |\mu_i| + |f_0 * \mu_i(u) - f_0 * \mu_0(u)|. \end{aligned}$$

The desired consequence follows from (2.10). □

2.3. **Radial Minkowski homomorphisms.** In order to prove Theorem 1.2, a characterization of radial Minkowski homomorphisms is crucial. We require the following characterization of linear operators on $C(S^{n-1})$. The theorem is a slight variation of a result by Dunkl [16], and its proof was given by Schuster [49, Theorem 5.1].

We say a linear map $\Psi : C(S^{n-1}) \to C(S^{n-1})$ is *monotone* if $f \leq g$ implies $\Psi f \leq \Psi g$ for every $f, g \in C(S^{n-1})$; equivalently, it maps nonnegative functions to nonnegative functions. It is $\mathrm{SO}(n)$-intertwining if for every $\vartheta \in \mathrm{SO}(n)$, we have

$$\Psi(\vartheta f) = \vartheta \Psi(f).$$

Here, for $g \in C(S^{n-1})$, the function ϑg is given by

$$\vartheta g(u) = g(\vartheta^{-1} u), \text{ for each } u \in S^{n-1}.$$

Theorem 2.2. *A map $\Psi : C(S^{n-1}) \to C(S^{n-1})$ is a monotone, linear map that intertwines $\mathrm{SO}(n)$ if and only if there is a measure $\mu \in \mathcal{M}(S^{n-1}, \widehat{e})$ such that*

$$\Psi f = f * \mu.$$

The following characterization of (r, q) radial Minkowski homomorphisms is a *slight* generalization (and modification) of Schuster [49, Theorem 5.2]. We include the proof here for completeness.

Theorem 2.3. *Let $r > 0$, $q \neq 0$, and $\Phi : \mathcal{S}_o^n \to \mathcal{S}_o^n$. The map Φ is an (r, q) radial Minkowski homomorphism if and only if there exists a nonzero $\mu \in \mathcal{M}(S^{n-1}, \widehat{e})$ such that*

$$\rho^q(\Phi K, \cdot) = \rho^r(K, \cdot) * \mu. \tag{2.11}$$

Proof. The verification of the "if" part is trivial and is therefore skipped. We only prove the "only if" part.

Note that since S^{n-1} is compact, for every $f \in C(S^{n-1})$, there exist positive functions $f_1, f_2 \in C(S^{n-1})$ such that $f = f_1 - f_2$. Define $L_1, L_2 \in \mathcal{S}_o^n$ such that $\rho_{L_i} = f_i^{\frac{1}{r}}$ on S^{n-1} for $i = 1, 2$. Note that for a given $f \in C(S^{n-1})$, the choice of L_1 and L_2 is not unique.

Define $\Psi : C(S^{n-1}) \to C(S^{n-1})$ by

$$\Psi(f) = \rho^q_{\Phi L_1} - \rho^q_{\Phi L_2}.$$

We first show that $\Psi(f)$ is well-defined; that is, it is independent of the choice of L_1 and L_2. Towards this end, assume there are $L_1', L_2' \in \mathcal{S}_o^n$ such that

$$f = f_1' - f_2' = \rho^r_{L_1'} - \rho^r_{L_2'}.$$

Then,

$$\rho_{L_1}^r - \rho_{L_2}^r = \rho_{L_1'}^r - \rho_{L_2'}^r,$$

or, equivalently,

$$L_1 \widetilde{+}_r L_2' = L_1' \widetilde{+}_r L_2.$$

Since Φ is an (r, q) radial Minkowski homomorphism, we have

$$\Phi(L_1) \widetilde{+}_q \Phi(L_2') = \Phi(L_1') \widetilde{+}_q \Phi(L_2),$$

or, equivalently,

$$\rho_{\Phi L_1}^q - \rho_{\Phi L_2}^q = \rho_{\Phi L_1'}^q - \rho_{\Phi L_2'}^q.$$

This shows that Ψ is well-defined.

The fact that Ψ intertwines $\mathrm{SO}(n)$ follows directly from the fact that Φ intertwines $\mathrm{SO}(n)$.

We claim that Ψ is continuous. Let $g_i, g \in C(S^{n-1})$ be such that g_i converges to g uniformly. Let $C > 0$ be such that $g_i, g \geq 1 - C$. Such a C exists since S^{n-1} is compact and the convergence of g_i is uniform. Set $Q \in \mathcal{S}_o^n$ so that $\rho_Q^r = C$. Hence, there exist $L_i, L \in \mathcal{S}_o^n$ such that

$$g_i = (g_i + C) - C = \rho_{L_i}^r - \rho_Q^r,$$
$$g = (g + C) - C = \rho_L^r - \rho_Q^r.$$

Since $r > 0$ and g_i converges to g uniformly, we have $L_i \to L$. Hence, by the continuity of Φ, we have

$$\Psi(g_i) = \rho_{\Phi L_i}^q - \rho_{\Phi Q}^q \to \rho_{\Phi L}^q - \rho_{\Phi Q}^q = \Psi(g)$$

uniformly on S^{n-1}. Here, we used the fact that since $\Phi L \in \mathcal{S}_o^n$, the functions $\rho_{\Phi L_i}$ are uniformly bounded away from 0.

We now show that Ψ is additive; that is, $\Psi(f+g) = \Psi(f) + \Psi(g)$. Suppose $L_1, L_2, K_1, K_2 \in \mathcal{S}_o^n$ are such that

$$f = \rho_{L_1}^r - \rho_{L_2}^r,$$
$$g = \rho_{K_1}^r - \rho_{K_2}^r.$$

Then

$$f + g = \rho_{L_1 \widetilde{+}_r K_1}^r - \rho_{L_2 \widetilde{+}_r K_2}^r.$$

Hence, by the definition of Ψ and that Φ is an (r, q) radial Minkowski homomorphism, we have

$$\Psi(f+g) = \rho_{\Phi(L_1 \widetilde{+}_r K_1)}^q - \rho_{\Phi(L_2 \widetilde{+}_r K_2)}^q = \rho_{\Phi L_1}^q + \rho_{\Phi K_1}^q - \rho_{\Phi L_2}^q - \rho_{\Phi K_2}^q = \Psi(f) + \Psi(g).$$

The additivity of Ψ and that it is continuous now implies that it is a linear map.

To invoke Theorem 2.2, it remains to check that Ψ is monotone. Let $f \in C(S^{n-1})$ be a nonnegative continuous function. For each $\varepsilon > 0$, consider $f_\varepsilon = f + \varepsilon > 0$. Let $L_1, L_2 \in \mathcal{S}_o^n$ be such that

$$f_\varepsilon = \rho_{L_1}^r - \rho_{L_2}^r.$$

Since $f_\varepsilon > 0$, there exists $K \in \mathcal{S}_o^n$ such that

$$\rho_K^r = f_\varepsilon = \rho_{L_1}^r - \rho_{L_2}^r,$$

or, equivalently,

$$K \widetilde{+}_r L_2 = L_1.$$

Since Φ is an (r, q) radial Minkowski homomorphism, we have

$$\rho_{\Phi L_1}^q = \rho_{\Phi K}^q + \rho_{\Phi L_2}^q.$$

This implies that $\Psi(f_\varepsilon) = \rho^q_{\Phi K} > 0$. Now let $\varepsilon \to 0^+$. By continuity of Ψ, we have $\Psi(f) \geq 0$.

Now, Theorem 2.2 implies the existence of $\mu \in \mathcal{M}(S^{n-1}, \widehat{e})$ such that

$$\Psi f = f * \mu. \tag{2.12}$$

We are finally in the position to prove (2.11). Indeed, for every $L \in \mathcal{S}^n_o$, there exists $c > 0$ such that

$$f = \rho^r_L - \rho^r_{cB} > 0.$$

By (2.12) and the definition of Ψ, we have

$$\rho^q_{\Phi L} - \rho^q_{\Phi cB} = \Psi f = (\rho^r_L - c^r) * \mu.$$

Letting $c \to 0^+$ completes the proof. □

At times, we refer to the measure μ in Theorem 2.3 as the *generating measure* of the (r, q) radial Minkowski homomorphism Φ.

Remark 2.4. *Simply from the definition of radial Minkowski homomorphisms, it is clear that we can count among them the classical intersection body operator, the L_p intersection body operator for $0 \neq p < 1$ and its asymmetric version, and the polar L_p centroid body for $0 \neq p > -1$. Theorem 2.3 reveals that there are a lot more radial Minkowski homomorphisms, one for every $\mu \in \mathcal{M}(S^{n-1}, \widehat{e})$.*

3. Radial Minkowski homomorphisms and Φ-surface area measures

The following definition of Φ-surface area measure was given in the Introduction. We repeat it here for the convenience of the reader.

Definition 3.1 (L_p Φ-surface area measure)**.** *Let $r > 0, q \neq 0, p \in \mathbb{R}$ and $\Phi : \mathcal{S}^n_o \to \mathcal{S}^n_o$ be an (r, q) radial Minkowski homomorphism. For every $K \in \mathcal{K}^n_o$, its L_p* Φ-surface area measure, *denoted by $S_{\Phi,p}(K, \cdot)$, is a finite Borel measure on S^{n-1} given by*

$$S_{\Phi,p}(K, \omega) = \int_{\boldsymbol{\alpha}^*_K(\omega)} (\rho_K(u)u \cdot \alpha_K(u))^{-p} \rho^q_{\Phi\mathcal{T}\Phi K}(u) \rho^r_K(u) du.$$

The main purpose of this section is to prove Theorem 1.2, which reveals that the Φ-surface area measure naturally arises by properly differentiating the Φ-volume $V(\Phi\,\cdot)$. Compare this to the classical surface area measure appearing in (1.1), which arises by properly differentiating the volume $V(\cdot)$.

To prove Theorem 1.2, we need the following two lemmas.

Lemma 3.2. *Let $r > 0, q \neq 0$, and $\Omega \subset S^{n-1}$ be a closed set not contained in any closed hemisphere. Suppose $h_0, f : \Omega \to (0, \infty)$ are continuous functions. For sufficiently small $|t|$, define $h_t : \Omega \to (0, \infty)$ by*

$$\log h_t = \log h_0 + tf + o(t, \cdot),$$

where $o(t, \cdot)$ is a continuous function on Ω and $o(t, \cdot)/t$, as $t \to 0$, converges uniformly to 0 on Ω. Write $K_t = [h_t]$. Let $\mu \in \mathcal{M}(S^{n-1}, \widehat{e})$. There exists $M > 0$ (independent of t and $u \in S^{n-1}$) such that

(1) for $\mathcal{H}^{n-1}$ almost all $u \in S^{n-1}$, we have

$$\lim_{t\to 0} \frac{\rho^r_{K_t}(u) - \rho^r_{K_0}(u)}{t} = r\rho^r_{K_0}(u) f(\alpha_{K_0}(u)); \tag{3.1}$$

(2) for all t with $|t|$ sufficiently small and $u \in S^{n-1}$,

$$\left|\frac{\rho^r_{K_t}(u) - \rho^r_{K_0}(u)}{t}\right| \le M, \tag{3.2}$$

and

$$\left|\frac{(\rho^r_{K_t} * \mu)^{\frac{n}{q}} - (\rho^r_{K_0} * \mu)^{\frac{n}{q}}}{t}\right| \le M. \tag{3.3}$$

Proof. Equation (3.1) is due to Huang-Lutwak-Yang-Zhang [26, Lemma 4.3], with an application of the chain rule. Equation (3.2) is also due to Huang-Lutwak-Yang-Zhang, which can also be deduced directly from [26, Lemma 4.2]. Thus, we only need to show (3.3).

Note that there exists $c > 0$ such that for all sufficiently small $|t|$, we have

$$\frac{1}{c} \le \rho^r_{K_t} \le c$$

and consequently,

$$\frac{1}{c}|\mu| \le \rho^r_{K_t} * \mu \le c|\mu|.$$

Therefore, we may use the mean value theorem and (3.2) to deduce the existence of ξ between $\rho^r_{K_t} * \mu$ and $\rho^r_{K_0} * \mu$ ($\xi \in (\frac{1}{c}|\mu|, c|\mu|)$ consequently) such that

$$\left|\frac{(\rho^r_{K_t} * \mu)^{\frac{n}{q}} - (\rho^r_{K_0} * \mu)^{\frac{n}{q}}}{t}\right| = \left|\frac{n}{q}\right| \xi^{\frac{n}{q}-1} \left|\left(\frac{\rho^r_{K_t}(u) - \rho^r_{K_0}(u)}{t}\right) * \mu\right| \le \left|\frac{n}{q}\right| \xi^{\frac{n}{q}-1} M|\mu|.$$

This proves (3.3) by possibly replacing M by $\left|\frac{n}{q}\right| \xi^{\frac{n}{q}-1} M|\mu|$ if the latter is larger. $\square$

We remark that under the assumptions of Lemma 3.2, if $t > 0$ is sufficiently small, then $\rho_{K_t} > \rho_{K_0}$. Indeed, since f is a positive continuous function on S^{n-1} and $o(t,\cdot)/t$ converges uniformly to 0, we have $h_t > h_0 + ct$ on Ω for some $c > 0$. This implies $K_0 + ctB \subset K_t$, which proves the claim. Similarly, if $t < 0$ is sufficiently close to 0, then $\rho_{K_0} > \rho_{K_t}$. An immediate consequence is that for each nonzero $\mu \in \mathcal{M}(S^{n-1}, \widehat{e})$ and $t \neq 0$ sufficiently close to 0, we have $\rho^r_{K_t} * \mu \neq \rho^r_{K_0} * \mu$ everywhere on S^{n-1}.

Lemma 3.3. *Under the assumptions of Lemma 3.2, as $t \to 0$, we have*

$$\frac{(\rho^r_{K_t} * \mu)^{\frac{n}{q}} - (\rho^r_{K_0} * \mu)^{\frac{n}{q}}}{\rho^r_{K_t} * \mu - \rho^r_{K_0} * \mu} \to \frac{n}{q}(\rho^r_{K_0} * \mu)^{\frac{n}{q}-1}, \tag{3.4}$$

uniformly on S^{n-1} and consequently,

$$\left(\frac{(\rho^r_{K_t} * \mu)^{\frac{n}{q}} - (\rho^r_{K_0} * \mu)^{\frac{n}{q}}}{\rho^r_{K_t} * \mu - \rho^r_{K_0} * \mu}\right) * \mu \to \frac{n}{q}\left((\rho^r_{K_0} * \mu)^{\frac{n}{q}-1}\right) * \mu, \tag{3.5}$$

uniformly on S^{n-1}.

Proof. Note that by the definition of K_t, we have $\rho_{K_t}^r \to \rho_{K_0}^r$ uniformly on S^{n-1}. Consequently, we have $\rho_{K_t}^r * \mu \to \rho_{K_0}^r * \mu > 0$ uniformly on S^{n-1}. By the mean value theorem, there exists ξ between $\rho_{K_t}^r * \mu$ and $\rho_{K_0}^r * \mu$ such that

$$\frac{(\rho_{K_t}^r * \mu)^{\frac{n}{q}} - (\rho_{K_0}^r * \mu)^{\frac{n}{q}}}{\rho_{K_t}^r * \mu - \rho_{K_0}^r * \mu} = \frac{n}{q}\xi^{\frac{n}{q}-1}.$$

This, when combined with the fact that $\rho_{K_t}^r * \mu \to \rho_{K_0}^r * \mu$ uniformly on S^{n-1}, implies (3.4). Equation (3.5) follows directly from (3.4). □

We are ready to prove Theorem 1.2.

Proof of Theorem 1.2. By Theorem 2.3, there exists a nonzero $\mu \in \mathcal{M}(S^{n-1}, \widehat{e})$ such that

$$\rho_{\Phi K}^q = \rho_K^r * \mu, \qquad \text{for each } K \in \mathcal{S}_o^n.$$

We first prove the case $p = 0$. By polar coordinates and (2.9), we have

$$\begin{aligned}
& n\frac{V(\Phi K_t) - V(\Phi K_0)}{t} \\
&= \int_{S^{n-1}} \frac{(\rho_{K_t}^r * \mu)^{\frac{n}{q}} - (\rho_{K_0}^r * \mu)^{\frac{n}{q}}}{t} du \\
&= \int_{S^{n-1}} \left(\frac{(\rho_{K_t}^r * \mu)^{\frac{n}{q}} - (\rho_{K_0}^r * \mu)^{\frac{n}{q}}}{\rho_{K_t}^r * \mu - \rho_{K_0}^r * \mu} \right) \cdot \frac{\rho_{K_t}^r * \mu - \rho_{K_0}^r * \mu}{t} du \\
&= \int_{S^{n-1}} \left[\left(\frac{(\rho_{K_t}^r * \mu)^{\frac{n}{q}} - (\rho_{K_0}^r * \mu)^{\frac{n}{q}}}{\rho_{K_t}^r * \mu - \rho_{K_0}^r * \mu} \right) * \mu \right] \cdot \frac{\rho_{K_t}^r - \rho_{K_0}^r}{t} du.
\end{aligned}$$

Lemmas 3.2, 3.3, the dominated convergence theorem, now imply

$$\begin{aligned}
\lim_{t\to 0} n\frac{V(\Phi K_t) - V(\Phi K_0)}{t} &= \frac{nr}{q} \int_{S^{n-1}} \left[\left((\rho_{K_0}^r * \mu)^{\frac{n}{q}-1} \right) * \mu \right] \rho_{K_0}^r(u) f(\alpha_{K_0}(u)) du \\
&= \frac{nr}{q} \int_{S^{n-1}} \left[\rho_{\Phi K_0}^{n-q} * \mu \right] \rho_{K_0}^r(u) f(\alpha_{K_0}(u)) du \\
&= \frac{nr}{q} \int_{S^{n-1}} [\rho_{\mathcal{T}\Phi K_0}^r * \mu] \rho_{K_0}^r(u) f(\alpha_{K_0}(u)) du \\
&= \frac{nr}{q} \int_{S^{n-1}} \rho_{\Phi\mathcal{T}\Phi K_0}^q(u) \rho_{K_0}^r(u) f(\alpha_{K_0}(u)) du \\
&= \frac{nr}{q} \int_{\Omega} f(v) dS_{\Phi,0}(K_0, v),
\end{aligned}$$

where in the last line we used the definition of $S_{\Phi,0}(K_0, \cdot)$ and the fact that since $K_0 = [h_0, \Omega]$, we have $S_{\Phi,0}(K_0, S^{n-1} \setminus \Omega) = 0$ (a consequence of $S(K, S^{n-1} \setminus \Omega) = 0$).

We now prove the $p \neq 0$ case. Assume h_t is given by $h_t^p = h_0^p + tf + o(t, \cdot)$. Consequently,

$$\log h_t = \frac{1}{p} \log(h_0^p + tf + o(t, \cdot)) = \log h_0 + \frac{1}{p} \log\left(1 + \frac{f}{h_0^p} t + \frac{o(t, \cdot)}{h_0^p}\right) = \log h_0 + \frac{1}{p}\frac{f}{h_0^p} t + o(t, \cdot).$$

The desired consequence now follows directly from the $p = 0$ case and the definition of $S_{\Phi,p}(K_0, \cdot)$. □

It is immediate from the definition that $S_{\Phi,p}(K,\cdot)$ is absolutely continuous with respect to the surface area measure of K. If K is C^2 with everywhere positive Gauss curvature, then $S_{\Phi,p}$ is absolutely continuous with respect to the spherical Lebesgue measure:

$$\frac{dS_{\Phi,p}(K,v)}{dv} = h_K^{1-p}(v)\rho^q_{\Phi\mathcal{T}\Phi K}(\alpha^*_K(v))|\nabla h_K(v) + h_K(v)v|^{r-n}\det(\nabla^2 h_K(v) + h_K(v)I)dv.$$

Recall that $\det(\nabla^2 h_K(v) + h_K(v)I)$ gives the reciprocal Gauss curvature of K at the boundary point corresponding to the unit normal v. Therefore, from a geometric point of view, the Φ-surface area measure $S_{\Phi,p}(K,\cdot)$ reduces to a version of the Gauss curvature if K is C^2 with everywhere positive Gauss curvature.

We prove some useful properties of the Φ-surface area measure in the remainder of this section.

The following corollaries are immediate from Theorem 2.3.

Corollary 3.4. *Let $r > 0$ and $q \neq 0$. If $\Phi : \mathcal{S}^n_o \to \mathcal{S}^n_o$ is an (r,q) radial Minkowski homomorphism, then*

$$\Phi(cK) = c^{\frac{r}{q}}\Phi(K),$$

for all $c > 0$ and $K \in \mathcal{S}^n_o$.

Corollary 3.5. *Let $r > 0$, $q \neq 0$ and $p \in \mathbb{R}$. If $\Phi : \mathcal{S}^n_o \to \mathcal{S}^n_o$ is an (r,q) radial Minkowski homomorphism, then*

$$S_{\Phi,p}(cK,\cdot) = c^{-p+\frac{rn}{q}}S_{\Phi,p}(K,\cdot),$$

for all $c > 0$ and $K \in \mathcal{K}^n_o$.

Corollary 3.6. *Let $r > 0$, $q \neq 0$ and $p \in \mathbb{R}$. If $\Phi : \mathcal{S}^n_o \to \mathcal{S}^n_o$ is an (r,q) radial Minkowski homomorphism, then ΦK is origin-symmetric as long as K is origin-symmetric. As a consequence, if K is origin-symmetric, then $S_{\Phi,p}(K,\cdot)$ is an even measure.*

Proof. By Theorem 2.3, there exists $\mu \in \mathcal{M}(S^{n-1}, \widehat{e})$ such that

$$\begin{aligned}
\rho^q_{\Phi K}(-u) &= \rho^r_K * \mu(-u)\\
&= \int_{S^{n-1}} \rho^r_K(\vartheta_{-u}v)d\mu(v)\\
&= \int_{S^{n-1}} \rho^r_K(-\vartheta_u v)d\mu(v)\\
&= \int_{S^{n-1}} \rho^r_K(\vartheta_u v)d\mu(v)\\
&= \rho^q_{\Phi K}(u).
\end{aligned}$$

Here, when going from the second to the third line, we used the fact that the integration is independent of the choice of $\vartheta \in \mathrm{SO}(n)$ as long as $\vartheta\widehat{e} = -u$ and that $\vartheta_{-u}\widehat{e} = -\vartheta_u\widehat{e} = -u$; when going from the third to the fourth line, we used the origin-symmetry of K.

That $S_{\Phi,p}(K,\cdot)$ is an even measure follows from its definition. □

The measure $S_{\Phi,p}(K,\cdot)$ is weakly continuous both in Φ and in K.

Proposition 3.7. *Let $r > 0, q \neq 0, p \in \mathbb{R}$ and $\Phi_i : \mathcal{S}^n_o \to \mathcal{S}^n_o$ be a sequence of (r,q) radial Minkowski homomorphisms with associated generating measures μ_i. If μ_i converges weakly to μ_0 and $K_i \in \mathcal{K}^n_o$ converges in the Hausdorff metric to $K_0 \in \mathcal{K}^n_o$, then the measures $S_{\Phi_i,p}(K_i,\cdot)$ converge weakly to $S_{\Phi_0,p}(K_0,\cdot)$.*

Proof. By Lemma 2.1, we have $\rho_{K_i}^r * \mu_i \to \rho_{K_0}^r * \mu_0$ uniformly on S^{n-1}.

Since $K_0 \in \mathcal{K}_o^n$ and $K_i \to K_0$, there exists $c_0 > 0$ such that

$$\frac{1}{c_0} < \rho_{K_i}^r < c_0.$$

When this is combined with the fact that $\mu_i(S^{n-1}) \to \mu_0(S^{n-1})$, this implies the existence of $c_1 > 0$ such that

$$\frac{1}{c_1} < \rho_{K_i}^r * \mu_i < c_1.$$

The fact that $\rho_{K_i}^r * \mu_i \to \rho_{K_0}^r * \mu_0$ uniformly now implies that $\rho_{\Phi_i K_i} \to \rho_{\Phi_0 K_0}$ uniformly and the existence of $c_2 > 0$ such that $c_2^{-1} < \rho_{\Phi_i K_i} < c_2$. This further implies that $\rho_{\mathcal{T}\Phi_i K_i} \to \rho_{\mathcal{T}\Phi_0 K_0}$ uniformly and the existence of $c_3 > 0$ such that $c_3^{-1} < \rho_{\mathcal{T}\Phi_i K_i} < c_3$. By the same argument, we conclude that $\rho_{\Phi_i \mathcal{T}\Phi_i K_i} \to \rho_{\Phi_0 \mathcal{T}\Phi_0 K_0}$ uniformly and the existence of $c_4 > 0$ such that $c_4^{-1} < \rho_{\Phi_i \mathcal{T}\Phi_i K_i} < c_4$.

Let $f \in C(S^{n-1})$ be arbitrary. By the definition of $S_{\Phi_i,p}(K_i,\cdot)$, we have

$$\int_{S^{n-1}} f(v) dS_{\Phi_i,p}(K_i,v) = \int_{S^{n-1}} f(\alpha_{K_i}(u))(\rho_{K_i}(u)u\cdot\alpha_{K_i}(u))^{-p}\rho^q_{\Phi_i\mathcal{T}\Phi_iK_i}(u)\rho^r_{K_i}(u)du. \tag{3.6}$$

Since K_i and $\Phi_i\mathcal{T}\Phi_iK_i$ are uniformly bounded, the integrand in (3.6) is uniformly bounded. Therefore, we may use the bounded convergence theorem, the facts that $\rho_{\Phi_i\mathcal{T}\Phi_iK_i} \to \rho_{\Phi_0\mathcal{T}\Phi_0K_0}$ and that α_{K_i} converges pointwise to α_{K_0} almost everywhere to conclude that

$$\begin{aligned}\int_{S^{n-1}} f(v) dS_{\Phi_i,p}(K_i,v) &\to \int_{S^{n-1}} f(\alpha_{K_0}(u))(\rho_{K_0}(u)u\cdot\alpha_{K_0}(u))^{-p}\rho^q_{\Phi_0\mathcal{T}\Phi_0K_0}(u)\rho^r_{K_0}(u)du \\ &= \int_{S^{n-1}} f(v)dS_{\Phi_0,p}(K_0,v).\end{aligned}$$

This establishes weak convergence. □

4. Shape recovery using $S_{\Phi,p}(K,\cdot)$

Can $S_{\Phi,p}(K,\cdot)$ be used to recover the shape of K and can it be done uniquely?

The L_p Φ-Minkowski problem. Let $r > 0$, $p \in \mathbb{R}, q \neq 0$ and $\Phi : \mathcal{S}_o^n \to \mathcal{S}_o^n$ be an (r,q) radial Minkowski homomorphism. Suppose ν is a given finite Borel measure on S^{n-1}. What are the (necessary and sufficient) conditions on ν so that there exists $K \in \mathcal{K}_o^n$ such that

$$\nu = S_{\Phi,p}(K,\cdot)?$$

Provided that the above existence question can be answered (even partially), the uniqueness question immediately follows. If $K, L \in \mathcal{K}_o^n$ satisfy $S_{\Phi,p}(K,\cdot) = S_{\Phi,p}(L,\cdot)$, how are they related?

The current section contains a full resolution of the existence of the solution to the L_p Minkowski problem for Φ-surface area measure when the given measure ν is even and $\frac{nr}{q} \neq p > 0$. Note that we do not impose any condition on Φ except that it is an (r,q) radial Minkowski homomorphism. Throughout this section, we fix such parameters r, p, q and Φ. We will also write $\mu \in \mathcal{M}(S^{n-1}, \widehat{e})$ for the generating measure of Φ in the sense of Theorem 2.3.

Consider the optimization problem

$$\inf_K \left\{ \int_{S^{n-1}} h_K^p d\nu : V(\Phi K) = 1, K \in \mathcal{K}_e^n \right\}. \tag{4.1}$$

The fact that there exists at least one $K \in \mathcal{K}_e^n$ such that $V(\Phi K) = 1$ comes from the fact that $V(\Phi\,\cdot)$ is homogeneous of degree $\frac{rn}{q}$ (see Corollary 3.4).

Lemma 4.1. *Let ν be an even finite Borel measure on S^{n-1}. If $K_0 \in \mathcal{K}_e^n$ satisfies $V(\Phi K_0) = 1$ and*

$$\int_{S^{n-1}} h_{K_0}^p d\nu = \inf_K \left\{ \int_{S^{n-1}} h_K^p d\nu : V(\Phi K) = 1, K \in \mathcal{K}_e^n \right\}, \tag{4.2}$$

then

$$\nu = S_{\Phi,p}(Q_0, \cdot),$$

where

$$Q_0 = \left(\frac{1}{n} \int_{S^{n-1}} h_{K_0}^p d\nu \right)^{\frac{q}{nr-pq}} K_0. \tag{4.3}$$

Proof. Consider the following relaxed optimization problem

$$\inf_h \left\{ \int_{S^{n-1}} h^p d\nu : V(\Phi[h]) = 1, h \in C_e^+(S^{n-1}) \right\}. \tag{4.4}$$

We claim that h_{K_0} is a minimizer of (4.4). Indeed, suppose $h \in C_e^+(S^{n-1})$ is such that $V(\Phi[h]) = 1$. Then by the definition of the Wulff shape, $p > 0$, and (4.2), we have

$$\int_{S^{n-1}} h^p d\nu \geq \int_{S^{n-1}} h_{[h]}^p d\nu \geq \int_{S^{n-1}} h_{K_0}^p d\nu.$$

This shows that h_{K_0} is a minimizer of (4.4).

Let $f \in C_e^+(S^{n-1})$ be arbitrary. For sufficiently small $|t|$, consider $h_t : S^{n-1} \to (0, \infty)$ given by

$$h_t^p = h_{K_0}^p + tf$$

and write $K_t = [h_t]$. Let $\lambda(t) > 0$ be such that $V(\Phi(\lambda(t)K_t)) = 1$, or, equivalently

$$\lambda(t) = V(\Phi K_t)^{-\frac{q}{rn}}.$$

It follows from Theorem 1.2 that

$$\lambda'(0) = -\frac{1}{pn} \int_{S^{n-1}} f(v) dS_{\Phi,p}(K_0, v).$$

Note that $\lambda(0) = 1$.

Consider $g_t = \lambda(t)h_t$. Note that by definition of $\lambda(t)$, we have $V(\Phi[g_t]) = 1$. Therefore, since h_{K_0} is a minimizer, we have

$$\begin{aligned}
0 &= \frac{d}{dt}\bigg|_{t=0} \int_{S^{n-1}} g_t^p d\nu \\
&= \frac{d}{dt}\bigg|_{t=0} \lambda(t)^p \int_{S^{n-1}} h_t^p d\nu \\
&= p\lambda(0)^{p-1}\lambda'(0) \int_{S^{n-1}} h_{K_0}^p d\nu + \lambda(0)^p \int_{S^{n-1}} f d\nu \\
&= -\frac{1}{n} \int_{S^{n-1}} h_{K_0}^p d\nu \int_{S^{n-1}} f(v) dS_{\Phi,p}(K_0, v) + \int_{S^{n-1}} f d\nu \\
&= -\int_{S^{n-1}} f(v) dS_{\Phi,p}(Q_0, v) + \int_{S^{n-1}} f d\nu.
\end{aligned}$$

Here, we use Corollary 3.5 in the last line and note that $p \neq \frac{nr}{q}$ guarantees that the power in (4.3) is defined.

Note that we established that for every positive, even, continuous function on S^{n-1},

$$\int_{S^{n-1}} f(v) dS_{\Phi,p}(Q_0, v) = \int_{S^{n-1}} f d\nu. \tag{4.5}$$

Since every even continuous function on S^{n-1} can be expressed as the difference of two positive, even continuous functions, we conclude that (4.5) holds for $f \in C_e(S^{n-1})$. Using the fact that both ν and $S_{\Phi,p}(Q_0, \cdot)$ are even (see Corollary 3.6), we conclude that

$$\nu = S_{\Phi,p}(Q_0, \cdot).$$

$\square$

Consider

$$\Xi = \{\xi \text{ is a proper subspace of } \mathbb{R}^n : \xi \neq \widehat{e}^{\perp}, \widehat{e} \notin \xi\}.$$

We will show that zonal measures have zero concentration on subspaces in Ξ.

Lemma 4.2. *Let $n \geq 3$ and $\mu \in \mathcal{M}(S^{n-1}, \widehat{e})$. Then for every $\xi \in \Xi$, we have $\mu(\xi \cap S^{n-1}) = 0$.*

Proof. Observe first that since $\mathrm{SO}(n-1)$ fixes $\widehat{e}$, if $\xi \in \Xi$, then $\vartheta\xi \in \Xi$ for every $\vartheta \in \mathrm{SO}(n-1)$.

We do an induction on the dimension of the subspaces in Ξ.

If $\xi \in \Xi$ is such that $k = \dim \xi = 1$, then $\xi \cap S^{n-1}$ contains two points. It is simple to see that since $\widehat{e} \notin \xi$, there exists infinitely many $\vartheta \in \mathrm{SO}(n-1)$ such that $\big(\vartheta(\xi \cap S^{n-1})\big) \cap \big(\xi \cap S^{n-1}\big) = \emptyset$. Since μ is zonal, if $\mu(\xi \cap S^{n-1}) > 0$, countable additivity would imply μ is an infinite measure, which is a contradiction.

Let us assume $\mu(\xi \cap S^{n-1}) = 0$ for all subspaces $\xi \in \Xi$ of dimensions between 1 and $k-1$. We will prove that the statement holds for k-dimensional subspaces.

Towards this end, assume $\xi \in \Xi$ is a k-dimensional subspace for some $k \leq n-1$. Let $j > 0$ be any arbitrary integer and $\vartheta_1, \dots, \vartheta_j \in \mathrm{SO}(n-1)$ be such that $\xi, \vartheta_1\xi, \dots, \vartheta_j\xi$ are distinct subspaces. For notational simplicity, let ϑ_0 be the identity map so that $\xi = \vartheta_0\xi$. (The existence of ϑ_i follows from the fact that $\xi \neq \widehat{e}^{\perp}$.) Note that for any $l \neq m$, we have $\vartheta_l\xi \cap \vartheta_m\xi$ is either the zero subspace, or a proper subspace of dimension at most $k-1$. Note also that since $\vartheta_l\xi, \vartheta_m\xi \in \Xi$, we have $\widehat{e} \notin \vartheta_l\xi \cap \vartheta_m\xi$ and consequently, if $\vartheta_l\xi \cap \vartheta_m\xi$ is not the zero subspace, then $\vartheta_l\xi \cap \vartheta_m\xi \in \Xi$. By the inductive hypothesis, we have $\mu\Big(\big(\vartheta_l\xi \cap \vartheta_m\xi\big) \cap S^{n-1}\Big) = 0$. Now, by countable additivity and the fact that μ is zonal, we have

$$|\mu| \geq \mu\left(\bigcup_{i=0}^{j} \vartheta_i\xi \cap S^{n-1}\right) = \sum_{i=0}^{j} \mu(\vartheta_i\xi \cap S^{n-1}) = (j+1)\mu(\xi \cap S^{n-1}).$$

Since j is an arbitrarily chosen positive integer, we conclude that $\mu(\xi \cap S^{n-1}) = 0$. $\square$

The following lemma is trivial.

Lemma 4.3. *Let μ be a finite Borel measure on S^1. Then there exists at most countably many 1-dimensional subspaces $\xi \subset \mathbb{R}^2$ such that $\mu(\xi \cap S^1) > 0$.*

Proof. Let $i > 0$ be arbitrary and consider

$$\Lambda_i = \left\{ \xi \subset \mathbb{R}^2 : \dim \xi = 1, \mu(\xi \cap S^1) > \frac{1}{i} \right\}.$$

Note that if $\xi_1, \xi_2 \in \Lambda_i$ and $\xi_1 \neq \xi_2$, then $(\xi_1 \cap S^1) \cap (\xi_2 \cap S^1) = \emptyset$. Hence, there exists at most finitely many elements in Λ_i, for, if there were infinitely many, by finite additivity of the measure, the measure μ would be an infinite measure.

Note that

$$\{\xi \subset \mathbb{R}^2 : \dim \xi = 1, \mu(\xi \cap S^1) > 0\} = \bigcup_{i=1}^{\infty} \Lambda_i.$$

This completes the proof. □

We also require the following lemma.

Lemma 4.4. *Let $K_i \in \mathcal{K}_e^n$ and $M > 0$. If $K_i \subset MB$ and there exists $v_0 \in S^{n-1}$ such that $h_{K_i}(v_0) \to 0$, then for sufficiently large i, $V(\Phi K_i) \neq 1$.*

Proof. For simplicity, we will write $g_i = \rho_{K_i}^r * \mu$ where $\mu \in \mathcal{M}(S^{n-1}, \widehat{e})$ is the generating measure of Φ in the sense of Theorem 2.3.

We now pick an arbitrary $v_1 \in S^{n-1}$ such that $v_1 \neq v_0$ and

$$\mu(S^{n-1} \cap v_1^{\perp}) = 0. \tag{4.6}$$

For $n \geq 3$, the existence of such v_1 is guaranteed by Lemma 4.2, whereas for $n = 2$, the existence is guaranteed by Lemma 4.3. Let $\vartheta \in \mathrm{SO}(n)$ be such that $\vartheta v_1 = v_0$ (equivalently, $\vartheta(v_1^{\perp}) = v_0^{\perp}$).

For each $\beta > 0$, define

$$\omega_\beta(v) = \{u \in S^{n-1} : |u \cdot v| \geq \beta\}.$$

In particular, note that $\vartheta[\omega_\beta(v)] = \omega_\beta(\vartheta v)$.

By the definition of the support function, if $u \in \omega_\beta(v_0)$, then

$$\rho_{K_i}(u)\beta \leq |\rho_{K_i}(u) u \cdot v_0| \leq h_{K_i}(v_0). \tag{4.7}$$

Hence,

$$\begin{aligned} g_i(\vartheta \widehat{e}) &= \rho_{K_i}^r * \mu(\vartheta \widehat{e}) \\ &= \int_{S^{n-1}} \rho_{K_i}^r(\vartheta u) d\mu(u) \\ &= \int_{\omega_\beta(v_1)} + \int_{S^{n-1} \setminus \omega_\beta(v_1)} \rho_{K_i}^r(\vartheta u) d\mu(u). \end{aligned} \tag{4.8}$$

Note that if $u \in \omega_\beta(v_1)$, then $\vartheta u \in \omega_\beta(\vartheta v_1) = \omega_\beta(v_0)$. Therefore, combining (4.7) and (4.8), we have

$$g_i(\vartheta \widehat{e}) \leq \left(\frac{h_{K_i}(v_0)}{\beta} \right)^r |\mu| + M^r \mu(S^{n-1} \setminus \omega_\beta(v_1)).$$

Let $\varepsilon > 0$ be arbitrary. Since μ is a finite measure and the set $S^{n-1} \setminus \omega_\beta(v_1)$ decreases to $v_1^{\perp}$ as $\beta \to 0^+$, we conclude from (4.6) that

$$\mu(S^{n-1} \setminus \omega_\beta(v_1)) \to \mu(v_1^{\perp} \cap S^{n-1}) = 0.$$

Hence, there exists $\beta_0 > 0$ sufficiently small so that

$$g_i(\widehat{\vartheta e}) \leq \left(\frac{h_{K_i}(v_0)}{\beta_0}\right)^r |\mu| + \varepsilon/2.$$

Now, using the fact that $h_{K_i}(v_0) \to 0$ as $i \to \infty$, we conclude the existence of $i_0 > 0$ such that for all $i > i_0$,

$$g_i(\widehat{\vartheta e}) < \varepsilon. \tag{4.9}$$

On the other hand, from the definition of g_i and the fact that $\rho_{K_i} \leq M$, we have

$$g_i(u) \leq M^r |\mu|, \qquad \forall u \in S^{n-1}. \tag{4.10}$$

We shall now complete the proof based on a discussion of the sign of q.

Case 1: if $q > 0$. In this case, from the definition of g_i, Theorem 2.3, and (4.9), we have

$$\rho_{\Phi K_i}(\widehat{\vartheta e}) = g_i(\widehat{\vartheta e})^{\frac{1}{q}} < \varepsilon^{\frac{1}{q}}. \tag{4.11}$$

Note that by Corollary 3.6, we have $\rho_{\Phi K_i}(-\widehat{\vartheta e}) < \varepsilon^{\frac{1}{q}}$. Similarly, using (4.10), we get

$$\rho_{\Phi K_i}(u) \leq (M^r |\mu|)^{\frac{1}{q}}, \qquad \forall u \in S^{n-1}. \tag{4.12}$$

Combining (4.11) and (4.12), we have

$$V(\Phi K_i) \leq 2\varepsilon^{\frac{1}{q}} (M^r |\mu|)^{\frac{n-1}{q}} \omega_{n-1}.$$

Since $\varepsilon > 0$ is arbitrary, we may take it sufficiently small so that for $i > i_0$,

$$V(\Phi K_i) \leq 2\varepsilon^{\frac{1}{q}} (M^r |\mu|)^{\frac{n-1}{q}} \omega_{n-1} < 1.$$

This completes the proof in the case of $q > 0$.

Case 2: if $q < 0$. In this case, from the definition of g_i, Theorem 2.3, and (4.9), we have

$$\rho_{\Phi K_i}(\widehat{\vartheta e}) = g_i(\widehat{\vartheta e})^{\frac{1}{q}} > \varepsilon^{\frac{1}{q}}. \tag{4.13}$$

Note that by Corollary 3.6, we have $\rho_{\Phi K_i}(-\widehat{\vartheta e}) > \varepsilon^{\frac{1}{q}}$. Similarly, using (4.10), we get

$$\rho_{\Phi K_i}(u) \geq (M^r |\mu|)^{\frac{1}{q}}, \qquad \forall u \in S^{n-1}. \tag{4.14}$$

Combining (4.13) and (4.14), we have

$$V(\Phi K_i) \geq \frac{2}{n}\varepsilon^{\frac{1}{q}} (M^r |\mu|)^{\frac{n-1}{q}} \omega_{n-1}.$$

Since $\varepsilon > 0$ is arbitrary, we may take it sufficiently small so that

$$V(\Phi K_i) \geq \frac{2}{n}\varepsilon^{\frac{1}{q}} (M^r |\mu|)^{\frac{n-1}{q}} \omega_{n-1} > 1.$$

This completes the proof in the case of $q < 0$. $\square$

Theorem 4.5. *Let $r, p > 0$, $q \neq 0$ with $\frac{nr}{q} \neq p$, $\Phi : \mathcal{S}_o^n \to \mathcal{S}_o^n$ be an (r, q) radial Minkowski homomorphism, and ν be a nonzero, finite, even Borel measure on S^{n-1}. There exists $K_0 \in \mathcal{K}_e^n$ such that $S_{\Phi,p}(K_0, \cdot) = \nu$ if and only if ν is not concentrated in any great subspheres.*

Proof. The "only if" part follows directly from the definition of $S_{\Phi,p}$ and that K_0 contains the origin as an interior point. Therefore, only the "if" part requires a proof.

Towards this end, let $K_i \in \mathcal{K}_e^n$ be a minimizing sequence to the optimization problem (4.1). In particular, this means $V(\Phi K_i) = 1$. Note that since there exists at least one $Q \in \mathcal{K}_e^n$ with $V(\Phi Q) = 1$, there exists $c_0 > 0$ such that

$$\int_{S^{n-1}} h_{K_i}^p d\nu < c_0.$$

Let $M_i = \max_{u\in S^{n-1}} \rho_{K_i}(u) = \rho_{K_i}(u_i)$. The existence of $u_i \in S^{n-1}$ comes from the fact that ρ_{K_i} is a continuous function on the compact set S^{n-1}. By the fact that K_i is origin-symmetric, and the definition of the support function, we have

$$M_i^p \int_{S^{n-1}} |u_i \cdot v|^p d\nu(v) \le \int_{S^{n-1}} h_{K_i}^p d\nu < c_0. \tag{4.15}$$

Note that the function

$$u \mapsto \int_{S^{n-1}} |u \cdot v|^p d\nu(v)$$

is a continuous function on S^{n-1}. Moreover, since ν is not concentrated in any great subspheres, this function is everywhere positive. Hence, there exists $c_1 > 0$ such that

$$\int_{S^{n-1}} |u_i \cdot v|^p d\nu(v) \ge c_1. \tag{4.16}$$

Combining (4.16) and (4.15), we have

$$M_i \le \left(\frac{c_0}{c_1}\right)^{\frac{1}{p}}.$$

Set $M = \left(\frac{c_0}{c_1}\right)^{\frac{1}{p}} > 0$. By the definition of M_i, we conclude that

$$K_i \subset MB.$$

Using Blaschke's selection theorem, by possibly taking a subsequence, we conclude that K_i converges in the Hausdorff metric to some K_0 where K_0 is a convex, compact set.

We now show that K_0 must contain the origin as an interior point. If not, then there exists $v_0 \in S^{n-1}$ such that $h_{K_0}(v_0) = 0$. Since $K_i \to K_0$, we have $\lim_{i\to\infty} h_{K_i}(v_0) = 0$. Lemma 4.4 now implies that for sufficiently large i, we have $V(\Phi K_i) \ne 1$. This is a contradiction.

By the continuity of the functional

$$K \mapsto \int_{S^{n-1}} h_K^p d\nu$$

and the functional $V(\Phi K)$ in K with respect to the Hausdorff metric, we conclude that K_0 is a minimizer of the optimization problem (4.1). Lemma 4.1 now implies the desired conclusion. □

The following Brunn-Minkowski-type inequality holds.

Theorem 4.6. *Let $0 < r < p$, $q \ne 0$, and $\Phi : \mathcal{S}_o^n \to \mathcal{S}_o^n$ be an (r, q) radial Minkowski homomorphism. Assume $q < 0$, or else $q > 0$ and $nr \le pq$. Then for every $K, L \in \mathcal{K}_o^n$ and $\lambda \in (0, 1)$, we have*

$$V(\Phi((1-\lambda)\cdot K +_p \lambda \cdot L))^{\frac{pq}{rn}} \ge (1-\lambda)V(\Phi K)^{\frac{pq}{rn}} + \lambda V(\Phi L)^{\frac{pq}{rn}}. \tag{4.17}$$

Moreover, if Φ is strictly monotone, then equality holds in (4.17) *if and only if K and L are dilates of each other.*

Proof. We prove (4.17) first under the additional assumption that $V(\Phi K) = V(\Phi L) = 1$.

By Theorem 2.3, there exists $\mu \in \mathcal{M}(S^{n-1}, \widehat{e})$ such that $\rho^q_{\Phi Q} = \rho^r_Q * \mu$, for every $Q \in \mathcal{S}^n_o$.

Case 1: when $q < 0$.

In this case, it is sufficient to show that $V(\Phi((1-\lambda)\cdot K +_p \lambda \cdot L)) \leq 1$. Thus, by (2.6) and the fact that $q < 0$, (2.8), that $\frac{p}{r} > 1$ together with the fact that $q < 0$, (2.8) again, the fact that $t^{\frac{rn}{pq}}$ is a convex function on $(0, \infty)$, we have for every $u \in S^{n-1}$,

$$
\begin{aligned}
\rho^n_{\Phi((1-\lambda)\cdot K +_p \lambda \cdot L)}(u) &= \left[\rho^r_{(1-\lambda)\cdot K +_p \lambda \cdot L} * \mu\right]^{\frac{n}{q}}(u) \\
&\leq \left[\rho^r_{(1-\lambda)\cdot K \widetilde{+}_p \lambda \cdot L} * \mu\right]^{\frac{n}{q}}(u) \\
&= \left[((1-\lambda)\rho^p_K + \lambda \rho^p_L)^{\frac{r}{p}} * \mu\right]^{\frac{n}{q}}(u) \\
&= \left[\int_{S^{n-1}} ((1-\lambda)\rho^p_K(\vartheta_u v) + \lambda \rho^p_L(\vartheta_u v))^{\frac{r}{p}}\, d\mu(v)\right]^{\frac{p}{r}\frac{rn}{pq}} \\
&\leq \left[(1-\lambda)\left(\int_{S^{n-1}} \rho^r_K(\vartheta_u v) d\mu(v)\right)^{\frac{p}{r}} + \lambda \left(\int_{S^{n-1}} \rho^r_L(\vartheta_u v) d\mu(v)\right)^{\frac{p}{r}}\right]^{\frac{rn}{pq}} \\
&= \left[(1-\lambda)\left(\rho^r_K * \mu\right)^{\frac{p}{r}} + \lambda \left(\rho^r_L * \mu\right)^{\frac{p}{r}}\right]^{\frac{rn}{pq}}(u) \\
&\leq (1-\lambda)\left(\rho^r_K * \mu\right)^{\frac{n}{q}}(u) + \lambda \left(\rho^r_L * \mu\right)^{\frac{n}{q}}(u) \\
&= (1-\lambda)\rho^n_{\Phi K}(u) + \lambda \rho^n_{\Phi L}(u).
\end{aligned}
\tag{4.18}
$$

Integrating (4.18) over $u \in S^{n-1}$ and using the fact that $V(\Phi K) = V(\Phi L) = 1$, we conclude that $V(\Phi((1-\lambda)\cdot K +_p \lambda \cdot L)) \leq 1$.

Case 2: when $q > 0$ and $nr \leq pq$. In this case, it is sufficient to show that $V(\Phi((1-\lambda)\cdot K +_p \lambda \cdot L)) \geq 1$. The proof is analogous. By (2.6), (2.8), that $\frac{p}{r} > 1$, (2.8) again, the fact that $nr \leq pq$

which implies that $t^{\frac{rn}{pq}}$ is a concave function on $(0,\infty)$, for every $u\in S^{n-1}$, we have

$$\begin{aligned}
\rho^n_{\Phi((1-\lambda)\cdot K+_p\lambda\cdot L)}(u) &= \left[\rho^r_{(1-\lambda)\cdot K+_p\lambda\cdot L}*\mu\right]^{\frac{n}{q}}(u)\\
&\geq \left[\rho^r_{(1-\lambda)\cdot K\widetilde{+}_p\lambda\cdot L}*\mu\right]^{\frac{n}{q}}(u)\\
&= \left[((1-\lambda)\rho_K^p+\lambda\rho_L^p)^{\frac{r}{p}}*\mu\right]^{\frac{n}{q}}(u)\\
&= \left[\int_{S^{n-1}}((1-\lambda)\rho_K^p(\vartheta_u v)+\lambda\rho_L^p(\vartheta_u v))^{\frac{r}{p}}\,d\mu(v)\right]^{\frac{p}{r}\frac{rn}{pq}}\\
&\geq \left[(1-\lambda)\left(\int_{S^{n-1}}\rho_K^r(\vartheta_u v)d\mu(v)\right)^{\frac{p}{r}}+\lambda\left(\int_{S^{n-1}}\rho_L^r(\vartheta_u v)d\mu(v)\right)^{\frac{p}{r}}\right]^{\frac{rn}{pq}}\\
&= \left[(1-\lambda)\left(\rho_K^r*\mu\right)^{\frac{p}{r}}+\lambda\left(\rho_L^r*\mu\right)^{\frac{p}{r}}\right]^{\frac{rn}{pq}}(u)\\
&\geq (1-\lambda)\left(\rho_K^r*\mu\right)^{\frac{n}{q}}(u)+\lambda\left(\rho_L^r*\mu\right)^{\frac{n}{q}}(u)\\
&= (1-\lambda)\rho^n_{\Phi K}(u)+\lambda\rho^n_{\Phi L}(u).
\end{aligned} \tag{4.19}$$

Integrating (4.19) over $u\in S^{n-1}$ and using the fact that $V(\Phi K)=V(\Phi L)=1$, we conclude that $V(\Phi((1-\lambda)\cdot K+_p\lambda\cdot L))\geq 1$.

For the general case, we apply the above result to $\widetilde{K}=\frac{K}{V(\Phi K)^{\frac{q}{rn}}}$, $\widetilde{L}=\frac{L}{V(\Phi L)^{\frac{q}{rn}}}$, and

$$\lambda=\frac{V(\Phi L)^{\frac{pq}{rn}}}{V(\Phi K)^{\frac{pq}{rn}}+V(\Phi L)^{\frac{pq}{rn}}}.$$

Note that

$$(1-\lambda)\cdot\widetilde{K}+_p\lambda\cdot\widetilde{L}=\frac{K+_pL}{\left(V(\Phi K)^{\frac{pq}{rn}}+V(\Phi L)^{\frac{pq}{rn}}\right)^{\frac{1}{p}}}.$$

Thus, in the case of $q<0$, we have

$$V\left(\Phi\left(\frac{K+_pL}{\left(V(\Phi K)^{\frac{pq}{rn}}+V(\Phi L)^{\frac{pq}{rn}}\right)^{\frac{1}{p}}}\right)\right)\leq 1,$$

or, equivalently,

$$V(\Phi(K+_pL))^{\frac{pq}{rn}}\geq V(\Phi K)^{\frac{pq}{rn}}+V(\Phi L)^{\frac{pq}{rn}}.$$

Inequality (4.17) now follows from this and the homogeneity of $V(\Phi Q)$ in Q.

The case $q>0$ and $nr\leq pq$ is analogous, and we omit the details.

If equality holds in (4.17), then the equality must hold in (4.18) and (4.19). In particular, it must be the case that

$$\Phi\left((1-\lambda)\cdot K+_p\lambda\cdot L\right)=\Phi\left((1-\lambda)\cdot K\widetilde{+}_p\lambda\cdot L\right).$$

Since Φ is strictly monotone, we conclude that $(1-\lambda)\cdot K+_p\lambda\cdot L=(1-\lambda)\cdot K\widetilde{+}_p\lambda\cdot L$. This implies that K and L are dilates of each other by the equality condition in (2.6). On the other hand, it is simple to verify that if K and L are dilates of each other, then equality holds in (4.17). $\square$

It is well known that a Brunn-Minkowski type inequality *with* equality condition implies the uniqueness of solutions to the corresponding Minkowski type problem. Hence, we state and prove the following uniqueness result.

Theorem 4.7. *Let $0 < r < p$, $q \neq 0$, and $\Phi : \mathcal{S}_o^n \to \mathcal{S}_o^n$ be a strictly monotone (r, q) radial Minkowski homomorphism. Assume $q < 0$, or else $q > 0$ and $nr < pq$. If $K, L \in \mathcal{K}_o^n$ satisfy*

$$S_{\Phi,p}(K, \cdot) = S_{\Phi,p}(L, \cdot), \tag{4.20}$$

then $K = L$.

Proof. Define $g : [0, \infty) \to \mathbb{R}$ by

$$g(t) = V(\Phi(K +_p t \cdot L))^{\frac{pq}{rn}} - V(\Phi K)^{\frac{pq}{rn}} - tV(\Phi L)^{\frac{pq}{rn}}.$$

It is simple to check that Theorem 4.6 implies that g is a nonnegative, concave function with $g(0) = 0$. In particular, this implies that the right derivative of g at 0 is nonnegative, or, equivalently, by Theorem 1.2,

$$V(\Phi K)^{\frac{pq}{rn}-1} \frac{1}{n} \int_{S^{n-1}} h_L^p(v) dS_{\Phi,p}(K, v) \geq V(\Phi L)^{\frac{pq}{rn}}. \tag{4.21}$$

Recall that since g is nonnegative, concave, and $g(0) = 0$, we have

$$0 \leq g(1) - g(0) \leq \left.\frac{d}{dt}\right|_{t=0^+} g(t).$$

Thus, if equality holds in (4.21), then $g(1) = 0$, which by Theorem 4.6, implies that K and L are dilates of each other.

We now prove that equality indeed holds in (4.21). Indeed, Theorem 1.2 and the homogeneity of $V(\Phi Q)$ in Q imply that for every $Q \in \mathcal{K}_o^n$,

$$\left.\frac{d}{dt}\right|_{t=0} (1+t)^{\frac{rn}{pq}} V(\Phi Q) = \left.\frac{d}{dt}\right|_{t=0} V(\Phi(Q +_p t \cdot Q)) = \frac{r}{pq} \int_{S^{n-1}} h_Q^p(v) dS_{\Phi,p}(Q, v),$$

or, equivalently,

$$V(\Phi Q) = \frac{1}{n} \int_{S^{n-1}} h_Q^p(v) dS_{\Phi,p}(Q, v).$$

This, (4.21), and (4.20) imply

$$V(\Phi K)^{\frac{pq}{rn}-1} V(\Phi L) = V(\Phi K)^{\frac{pq}{rn}-1} \frac{1}{n} \int_{S^{n-1}} h_L^p(v) dS_{\Phi,p}(K, v) \geq V(\Phi L)^{\frac{pq}{rn}}.$$

Reversing the role of K and L, we obtain the reverse inequality, and hence equality holds in (4.21). The fact that $K = L$ now follows from (4.20). $\square$

Declarations

Funding. Research of D. Xi was supported, in part, by the National Natural Science Foundation of China (NSFC) 12322103, and the Austrian Science Fund (FWF): 10.55776/P34446, 10.55776/P37030. Research of Y. Zhao was supported, in part, by the U.S. National Science Foundation DMS–2337630.

Data availability. Data sharing is not applicable to this article as no datasets were generated or analyzed during the current study.

Conflict of interest. On behalf of all authors, the corresponding author states that there is no conflict of interest.

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